NutriX – Invention Summary for Patent Review

Smart food-weighing system with hybrid AI identification and nutrition grounding

CONFIDENTIAL – draft of 31 October 2026. Applicant / inventors: Raj Kothari, Bhavya Sanghrajka, Harsh Maru, Chaitya Doshi.

Status key: **Built** = working in the prototype. **Designed** = specified, not yet built. **Partial** = part built.

1. Patentability ranking (strongest first)

#	Invention (what is claimed as new)	Status	Basis and main risk (Indian Patents Act)
1	Hybrid food identification on a weighing device: a first classifier is accepted only above a confidence gate and only if its class is not on a per-class distrust list; otherwise a local vision-language model labels the image, and the first model's guess is withheld for distrusted classes (anti-anchoring); label mapped to a canonical food; nutrients = measured weight × stored per-100 g values.	Built	Strongest. Technical effect: fewer unidentified/wrong captures, bounded latency, runs on a 6 GB GPU with no cloud call. Risk: crowded prior art; Sec. 3(k)/3(f). Claim the device plus the safeguards, not the software alone.
2	Plausibility-gated acceptance and repair of AI-generated nutrition values: energy checked against macro-nutrient masses, bounds test, repair-or-reject rule, and write-back of accepted values only, with provenance tags.	Built	Data-integrity control of a generative model feeding a persistent store. Risk: Sec. 3(k) (formula/rule). Claim as part of #1.
3	Per-user personalisation loop: model suggestion → one-tap confirmation → automatic bounding box → head-only retraining → small per-user detection head swapped onto one shared resident backbone.	Designed	Strong technical effect (many users, one backbone in GPU memory). Adapters are known in ML, so the loop and measured gains carry novelty. Must be built before the complete specification.
4	Scale hardware: floating circular platter on a single centre load cell inside a square enclosure; articulated camera arm fixed to the base (not the platter) giving a fixed top-down view.	Built (prototype, CAD)	Register the enclosure shape as a Design (fast, low cost, file before any disclosure). Utility claim only on a measured effect (camera mass/arm movement does not change the weight reading); Sec. 3(f) otherwise.
5	Closed-loop faithfulness auditor: compares model outputs with reference tables and, below a threshold, patches a runtime prompt registry; can trigger detector retraining.	Partial (registry built; auditor designed)	Plausible technical effect. Risk: Sec. 3(k). Build a minimal auditor first.
6	Weight-gated capture with delta guard and sequential-item weight attribution (stable-weight trigger; running total across items).	Partial (button capture, delta guard built; auto-trigger, multi-item designed)	Low novelty alone (stable-weight capture is common). Useful as a dependent feature of #1.
7	Weight-image synchronisation: fresh-frame capture after flushing the buffered frame, with weight and image sent in one streamed atomic request.	Built	Weak alone; include as a dependent feature of #1.
8	Single-writer macro synchronisation: one canonical payload from several writers, totals recomputed from source after insert and delete, fan-out to cache, event stream and live feed.	Built	Software architecture; Sec. 3(k). Better kept as copyright / trade secret.

Not expected to be patentable in India: meal-plan optimiser, health score, diet-rule tables (Sec. 3(m), 3(p)), barcode/label-reading chain, adaptive calorie targets, app features, and the use of YOLO, a vision-language model or OR-Tools as such. Protect these by copyright, trade secret, and the trademark; protect the enclosure by Design registration.

2. Complete feature inventory

Hardware

- **Weighing platform** – single centre load cell with 24-bit HX711 ADC under a raised circular platter in a square enclosure [Built]
- **Camera arm** – articulated column and tilted boom on the base giving a fixed top-down view of the platter [Built]
- **Controller and camera** – ESP32-S3 with OV2640 camera and live-view web server [Built]
- **Local display and controls** – OLED shows live weight and the result; capture button; power switch; rear service panel [Built]

Edge firmware

- **Button capture** – averaged weight reading, then a fresh camera frame (stale frame flushed) [Built]
- **Atomic upload** – weight and JPEG streamed in one multipart request with device token and device id; result shown on the OLED [Built]
- **Automatic stable-weight capture** – capture when readings stay stable for a set time, rejecting hand movement and settling [Designed]

Gateway and vision service

- **Gateway** – device-token authentication for scales, JWT for the app, request tracing [Built]
- **Food detector** – YOLOv8n, 123 produce and Indian-dish classes [Built]
- **Confidence gate and distrust list** – uncertain or known-confusable detections are routed to the vision-language model [Built]
- **Weight-change guard** – ignores changes under 0.5 g and negative tare events [Built]
- **Provenance and failure tags** – every log records which model identified it; a service outage logs zero nutrients, never an invented value [Built]
- **Pending-frame store** – low-confidence images kept for later labelling [Built]
- **Multi-item plate session** – running total across presses with per-item weight attribution [Partial]
- **One-tap confirmation** – model suggestion confirmed by the user, bounding box generated automatically [Partial]
- **Per-user head adapters and retraining** – shared backbone, small per-user heads, automatic retraining on repeated low confidence [Designed]

Local AI service

- **Local vision-language model** – Qwen3-VL-2B resident beside the detector on a 6 GB GPU; no cloud call [Built]
- **Structured answer** – label and per-100 g nutrients in one reply, with a name-only second pass [Built]
- **Energy-consistency guard** – range and macro-to-energy checks; repair or reject; flagged values never stored [Built]
- **Runtime prompt registry and audit log** – prompts editable without restart; every call logged [Built]

Knowledge layer

- **Tiered canonical lookup** – exact name, then alias, then trusted partial match [Built]
- **Self-growing food base** – accepted model values saved with provenance, aliases added; next lookup is a database hit [Built]
- **Composition data** – ICMR, USDA, UK and US tables, regional recipes and fast-food items [Built]

User and analytics service

- **Live macro synchronisation** – database trigger, cache, event stream and server-sent live dashboard [Built]
- **Dietary classifier** – vegan, vegetarian, Jain, Sattvic, halal, gluten-free and others, with substitutions [Built]
- **Meal planner** – OR-Tools mixed-integer plan per slot with a plan-candidate gate and soft fallback [Built]
- **Adaptive calorie targets** – weight trend and adherence adjust targets in bounded steps [Built]
- **Barcode and label reading** – product lookup chain, nutrition-label OCR, healthier and cheaper alternatives [Built]
- **Home-cooking builder** – household units (katori, tablespoon) converted to grams; community meal visibility per user [Built]
- **Recipe ranking, health score, explanations** – rule-based scoring and plain-language reasons [Built]

Agent layer (not yet built)

- **Orchestrator** – natural-language requests become typed tool calls and queries [Designed]
- **Guardian** – condition-specific thresholds on the live nutrient stream with escalation [Designed]
- **Clinical team** – diagnosis from logs, linear-programme corrective plan, report for dietitian sign-off [Designed]
- **Meta-auditor** – faithfulness score against reference tables; self-correction of prompts and detector [Designed]
- **Retry queue** – unresolved captures re-identified later and back-filled [Designed]

Client

- **Mobile app** – Flutter app with dashboard, scanning, recipes and biometric re-lock [Built]
- **Web dashboard** – live graph, captured images, device status, alerts [Partial]